

```
In [2]: import pandas as pd
import numpy as np
```

```
In [ ]:
```

```
In [3]: #Load dataset
dim_date = pd.read_csv(r"C:\Users\hongl\Downloads\64101194a2364 (2)\source-code\3_p
dim_hotel = pd.read_csv(r"C:\Users\hongl\Downloads\64101194a2364 (2)\source-code\3_
dim_room = pd.read_csv(r"C:\Users\hongl\Downloads\64101194a2364 (2)\source-code\3_p
fact_booking = pd.read_csv(r"C:\Users\hongl\Downloads\64101194a2364 (2)\source-code
fact_aggregated_booking = pd.read_csv(r"C:\Users\hongl\Downloads\64101194a2364 (2)\
new_data_august = pd.read_csv(r"C:\Users\hongl\Downloads\64101194a2364 (2)\source-c
```

```
In [4]: #profile the dataset
datasets = {
    'dim_dates': dim_date,
    "dim_hotels": dim_hotel,
    "dim_rooms": dim_room,
    "fact_bookings": fact_booking,
    "fact_aggregated_bookings": fact_aggregated_booking,
    "new_data_augusts": new_data_august
}
```

```
In [11]: for name, df in datasets.items():
    print('=' * 50)
    print(name)
    print("Shape:")
    print(df.shape)
    print(df.columns)
    print(df.dtypes)
    print(df.isnull().sum())
```

```

=====
dim_dates
Shape:
(92, 4)
Index(['date', 'mmm yy', 'week no', 'day_type'], dtype='object')
date          object
mmm yy        object
week no       object
day_type      object
dtype: object
date          0
mmm yy        0
week no       0
day_type      0
dtype: int64
=====

dim_hotels
Shape:
(25, 4)
Index(['property_id', 'property_name', 'category', 'city'], dtype='object')
property_id    int64
property_name  object
category       object
city           object
dtype: object
property_id    0
property_name  0
category       0
city           0
dtype: int64
=====

dim_rooms
Shape:
(4, 2)
Index(['room_id', 'room_class'], dtype='object')
room_id        object
room_class     object
dtype: object
room_id        0
room_class     0
dtype: int64
=====

fact_bookings
Shape:
(134590, 12)
Index(['booking_id', 'property_id', 'booking_date', 'check_in_date',
      'checkout_date', 'no_guests', 'room_category', 'booking_platform',
      'ratings_given', 'booking_status', 'revenue_generated',
      'revenue_realized'],
      dtype='object')
booking_id      object
property_id     int64
booking_date    object
check_in_date   object
checkout_date   object
no_guests       float64

```

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room_category      object
booking_platform    object
ratings_given       float64
booking_status      object
revenue_generated   int64
revenue_realized    int64
dtype: object
booking_id          0
property_id         0
booking_date        0
check_in_date       0
checkout_date       0
no_guests           3
room_category       0
booking_platform    0
ratings_given       77907
booking_status      0
revenue_generated   0
revenue_realized    0
dtype: int64

```

```

=====
fact_aggregated_bookings
Shape:
(9200, 5)
Index(['property_id', 'check_in_date', 'room_category', 'successful_bookings',
       'capacity'],
      dtype='object')

```

```

property_id         int64
check_in_date       object
room_category       object
successful_bookings int64
capacity            float64
dtype: object
property_id         0
check_in_date       0
room_category       0
successful_bookings 0
capacity            2
dtype: int64

```

```

=====
new_data_augusts
Shape:
(7, 13)
Index(['property_id', 'property_name', 'category', 'city', 'room_category',
       'room_class', 'check_in_date', 'mmm yy', 'week no', 'day_type',
       'successful_bookings', 'capacity', 'occ%'],
      dtype='object')

```

```

property_id         int64
property_name       object
category            object
city               object
room_category       object
room_class          object
check_in_date       object
mmm yy             object
week no            object

```

```

day_type          object
successful_bookings int64
capacity          int64
occ%              float64
dtype: object
property_id       0
property_name     0
category          0
city              0
room_category     0
room_class        0
check_in_date     0
mmm yy            0
week no           0
day_type          0
successful_bookings 0
capacity          0
occ%              0
dtype: int64

```

```

In [12]: #check duplicates
fact_booking['booking_id'].duplicated().sum()

```

```

Out[12]: np.int64(0)

```

```

In [15]: #missing value
fact_booking.isnull().sum()

```

```

Out[15]: booking_id          0
property_id          0
booking_date         0
check_in_date        0
checkout_date        0
no_guests            3
room_category        0
booking_platform     0
ratings_given       77907
booking_status       0
revenue_generated    0
revenue_realized     0
dtype: int64

```

```

In [18]: #check invalid guest counts
fact_booking[fact_booking['no_guests']<=0]

```

Out[18]:

		booking_id	property_id	booking_date	check_in_date	checkout_date	no_
0	May012216558RT11	16558	27-04-22	1/5/2022	2/5/2022		
3	May012216558RT14	16558	28-04-22	1/5/2022	2/5/2022		
17924	May122218559RT44	18559	12/5/2022	12/5/2022	14-05-22		
18020	May122218561RT22	18561	8/5/2022	12/5/2022	14-05-22		
18119	May122218562RT311	18562	5/5/2022	12/5/2022	17-05-22		
18121	May122218562RT313	18562	10/5/2022	12/5/2022	17-05-22		
56715	Jun082218562RT12	18562	5/6/2022	8/6/2022	13-06-22		
119765	Jul202219560RT220	19560	19-07-22	20-07-22	22-07-22		
134586	Jul312217564RT47	17564	30-07-22	31-07-22	1/8/2022		

In [19]: `print(len(fact_booking[fact_booking['no_guests']<=0]))`

9

In [25]: `# check revenue outlier
fact_booking[['revenue_generated', 'revenue_realized']].describe()`

Out[25]:

	revenue_generated	revenue_realized
count	1.345900e+05	1.345900e+05
mean	1.537805e+04	1.537805e+04
std	9.303604e+04	9.303604e+04
min	6.500000e+03	6.500000e+03
25%	9.900000e+03	9.900000e+03
50%	1.350000e+04	1.350000e+04
75%	1.800000e+04	1.800000e+04
max	2.856000e+07	2.856000e+07

In [26]: `fact_booking.sort_values(
 'revenue_realized',
 ascending = False
)[[
 'booking_id',
 'revenue_generated',
 'revenue_realized'
]].head(10)`

Out[26]:

	booking_id	revenue_generated	revenue_realized
111	May012216559RT32	28560000	28560
315	May012216562RT22	12600000	12600
129176	Jul282216562RT26	10000000	12600
2	May012216558RT13	9100000	9100
562	May012217559RT118	2000000	4420
66910	Jun152216559RT43	45220	18088
126791	Jul262216559RT44	45220	45220
30079	May212217563RT41	45220	45220
109188	Jul132217560RT45	45220	45220
9115	May072216559RT44	45220	18088

```
In [28]: # rule: successful_bookings <= capacity, unless allowed overbookings
fact_aggregated_booking[fact_aggregated_booking['successful_bookings']>fact_aggrega
```

Out[28]:

	property_id	check_in_date	room_category	successful_bookings	capacity
3	17558	1-May-22	RT1	30	19.0
12	16563	1-May-22	RT1	100	41.0
4136	19558	11-Jun-22	RT2	50	39.0
6209	19560	2-Jul-22	RT1	123	26.0
8522	19559	25-Jul-22	RT1	35	24.0
9194	18563	31-Jul-22	RT4	20	18.0

```
In [32]: # time formate
def parse_date_column(series):
    parsed = pd.to_datetime(
        series,
        dayfirst=True,
        format='mixed',
        errors='coerce'
    )
    return parsed
```

```
In [34]: fact_booking['booking_date']=(
    parse_date_column(fact_booking['booking_date']))
print(fact_booking['booking_date'].head(5))
```

```
0    2022-04-27
1    2022-04-30
2    2022-04-28
3    2022-04-28
4    2022-04-27
Name: booking_date, dtype: datetime64[ns]
```

```
In [37]: fact_booking['check_in_date']=(
        parse_date_column(fact_booking['check_in_date']))
        )
```

```
In [38]: fact_booking['checkout_date']=(
        parse_date_column(fact_booking['checkout_date']))
        )
```

```
In [39]: duplicate_booking_ids = fact_booking["booking_id"].duplicated().sum()

        print("Duplicate booking IDs:", duplicate_booking_ids)
```

Duplicate booking IDs: 0

```
In [40]: # rule : booking_date <= check_in_date < checkout_date
        invalid_booking_dates = fact_booking[
            fact_booking["booking_date"] >
            fact_booking["check_in_date"]
        ]

        print(
            "Booking after check-in:",
            len(invalid_booking_dates)
        )
```

Booking after check-in: 0

```
In [41]: invalid_checkout_dates = fact_booking[
        fact_booking["checkout_date"] <=
        fact_booking["check_in_date"]
    ]

    print(
        "Invalid checkout dates:",
        len(invalid_checkout_dates)
    )
```

Invalid checkout dates: 0

```
In [42]: #every booking belongs to a valid hotel.
        invalid_property_ids = fact_booking[
            ~fact_booking["property_id"].isin(
                dim_hotel["property_id"]
            )
        ]
```

```
In [43]: print(
        "Invalid property IDs:",
```

```
len(invalid_property_ids)
)
```

Invalid property IDs: 0

```
In [44]: invalid_room_ids = fact_booking[
~fact_booking["room_category"].isin(
    dim_room["room_id"]
)
]

print(
    "Invalid room categories:",
    len(invalid_room_ids)
)
```

Invalid room categories: 0

create new columns

```
In [47]: # 1. how far in advance do customers book
# formula ----Lead_time = check_in_date - booking_date
fact_booking["booking_lead_days"] = (
    fact_booking["check_in_date"]
    -
    fact_booking["booking_date"]
).dt.days
```

```
In [48]: fact_booking[
    [
        "booking_date",
        "check_in_date",
        "booking_lead_days"
    ]
].head()
```

```
Out[48]:
```

	booking_date	check_in_date	booking_lead_days
0	2022-04-27	2022-05-01	4
1	2022-04-30	2022-05-01	1
2	2022-04-28	2022-05-01	3
3	2022-04-28	2022-05-01	3
4	2022-04-27	2022-05-01	4

Length of staying

```
In [52]: fact_booking["length_of_staying"]=(fact_booking['checkout_date']-fact_booking['check_in_date']).dt.days
```

```
In [53]: fact_booking['length_of_staying'].head(5)
```

```
Out[53]: 0    1
          1    1
          2    3
          3    1
          4    1
          Name: length_of_staying, dtype: int64
```

```
In [55]: fact_booking['booking_lead_days'].describe()
```

```
Out[55]: count    134590.000000
          mean       3.708797
          std       4.877307
          min       0.000000
          25%       1.000000
          50%       3.000000
          75%       4.000000
          max      24.000000
          Name: booking_lead_days, dtype: float64
```

```
In [56]: fact_booking['length_of_staying'].describe()
```

```
Out[56]: count    134590.000000
          mean       2.374552
          std       1.733514
          min       1.000000
          25%       1.000000
          50%       2.000000
          75%       3.000000
          max       6.000000
          Name: length_of_staying, dtype: float64
```

```
In [60]: fact_booking[
          fact_booking["booking_lead_days"] < 0
        ]
fact_booking[
    fact_booking["length_of_staying"] <= 0
]
```

```
Out[60]:   booking_id  property_id  booking_date  check_in_date  checkout_date  no_guests  room_c
```

```
In [61]: # 0-3 days
          # 4-7 days
          # 8-14 days
          # 15-30 days
          # 31+ days
          bins = [-1, 3, 7, 14, 30, float("inf")]

          labels = [
              "0-3 days",
              "4-7 days",
              "8-14 days",
```

```
    "15-30 days",
    "31+ days"
]
```

```
In [65]: fact_booking["lead_time_group"] = pd.cut(
    fact_booking["booking_lead_days"],
    bins=bins,
    labels=labels
)
```

```
In [68]: fact_booking.to_csv(
    r"C:\Users\hongl\Downloads\64101194a2364 (2)\source-code\3_project_hospitality_
    index=False
)
```

```
In [69]: # check point
# =====
# FINAL DATA QUALITY VALIDATION
# =====

validation_results = {
    "duplicate_booking_ids":
        fact_booking["booking_id"]
        .duplicated()
        .sum(),

    "invalid_guest_counts":
        (fact_booking["no_guests"] <= 0)
        .sum(),

    "negative_revenue":
        (fact_booking["revenue_realized"] < 0)
        .sum(),

    "booking_after_checkin":
        (
            fact_booking["booking_date"]
            >
            fact_booking["check_in_date"]
        ).sum(),

    "invalid_checkout":
        (
            fact_booking["checkout_date"]
            <=
            fact_booking["check_in_date"]
        ).sum()
}

validation_df = pd.DataFrame(
    validation_results.items(),
    columns=[
        "validation_check",
        "failed_records"
    ]
)
```

```
)

print(validation_df)

validation_check  failed_records
0  duplicate_booking_ids          0
1  invalid_guest_counts           9
2      negative_revenue           0
3  booking_after_checkin           0
4      invalid_checkout           0
```

```
In [72]: # =====
# DERIVED COLUMNS
# =====

fact_booking["booking_lead_days"] = (
    fact_booking["check_in_date"]
    -
    fact_booking["booking_date"]
).dt.days

fact_booking["length_of_stay"] = (
    fact_booking["checkout_date"]
    -
    fact_booking["check_in_date"]
).dt.days
```

```
In [73]: fact_booking.dtypes
```

```
Out[73]: booking_id          object
property_id          int64
booking_date         datetime64[ns]
check_in_date        datetime64[ns]
checkout_date        datetime64[ns]
no_guests            float64
room_category        object
booking_platform      object
ratings_given        float64
booking_status        object
revenue_generated     int64
revenue_realized     int64
booking_lead_days     int64
length_of_staying     timedelta64[ns]
length_of_staying     int64
lead_time_group       category
length_of_stay        int64
dtype: object
```

```
In [76]: invalid_guests = fact_booking[
    fact_booking["no_guests"] <= 0
]
invalid_guests[
    [
        "booking_id",
        "property_id",
        "no_guests",
```

```

        "booking_status",
        "revenue_realized"
    ]
]

```

Out[76]:

	booking_id	property_id	no_guests	booking_status	revenue_realized
0	May012216558RT11	16558	-3.0	Checked Out	10010
3	May012216558RT14	16558	-2.0	Cancelled	3640
17924	May122218559RT44	18559	-10.0	No Show	20900
18020	May122218561RT22	18561	-12.0	Cancelled	3600
18119	May122218562RT311	18562	-6.0	Checked Out	16800
18121	May122218562RT313	18562	-4.0	Cancelled	5760
56715	Jun082218562RT12	18562	-17.0	Checked Out	6500
119765	Jul202219560RT220	19560	-1.0	Checked Out	13500
134586	Jul312217564RT47	17564	-4.0	Checked Out	38760

In [77]:

```

fact_booking.loc[
    fact_booking["no_guests"] <= 0,
    "no_guests"
] = pd.NA

```

In [78]:

```

fact_booking[fact_booking["no_guests"] <= 0]

```

Out[78]:

booking_id	property_id	booking_date	check_in_date	checkout_date	no_guests	room_c
------------	-------------	--------------	---------------	---------------	-----------	--------

In [79]:

```

fact_booking["no_guests"] = (
    fact_booking["no_guests"]
    .astype("Int64")
)

```

In [80]:

```

fact_booking["no_guests"].dtype

```

Out[80]: Int64Dtype()

In [81]:

```

missing_guest_counts = (
    fact_booking["no_guests"]
    .isna()
    .sum()
)

print(
    "Missing guest counts:",
    missing_guest_counts
)

```

Missing guest counts: 12

```
In [82]: fact_booking["guest_count_available"] = (  
        fact_booking["no_guests"].notna()  
        )
```

```
In [83]: fact_booking[fact_booking["guest_count_available"]].head(5)
```

```
Out[83]:
```

	booking_id	property_id	booking_date	check_in_date	checkout_date	no_guests
1	May012216558RT12	16558	2022-04-30	2022-05-01	2022-05-02	2
2	May012216558RT13	16558	2022-04-28	2022-05-01	2022-05-04	2
4	May012216558RT15	16558	2022-04-27	2022-05-01	2022-05-02	4
5	May012216558RT16	16558	2022-05-01	2022-05-01	2022-05-03	2
6	May012216558RT17	16558	2022-04-28	2022-05-01	2022-05-06	2

```
In [84]: fact_booking.drop(  
        columns=[  
            "length_of_staying",  
            "length_of_staying"  
        ],  
        inplace=True  
        )
```

```
In [86]: fact_booking["length_of_stay"].describe()
```

```
Out[86]: count    134590.000000  
mean         2.374552  
std          1.733514  
min          1.000000  
25%          1.000000  
50%          2.000000  
75%          3.000000  
max          6.000000  
Name: length_of_stay, dtype: float64
```

```
In [87]: invalid_stay = fact_booking[  
        fact_booking["length_of_stay"] <= 0  
        ]  
  
print(  
    "Invalid lengths of stay:",  
    len(invalid_stay)  
    )
```

Invalid lengths of stay: 0

```
In [88]: negative_lead_time = fact_booking[  
        fact_booking["booking_lead_days"] < 0  
        ]
```

```
print(
    "Negative lead-time records:",
    len(negative_lead_time)
)
```

Negative lead-time records: 0

```
In [90]: fact_booking["lead_time_group"] = (
    fact_booking["lead_time_group"]
    .astype("string")
)
```

```
In [91]: fact_booking["revenue_ratio"] = (
    fact_booking["revenue_generated"]
    /
    fact_booking["revenue_realized"]
)
```

```
In [92]: fact_booking[
    [
        "revenue_generated",
        "revenue_realized",
        "revenue_ratio"
    ]
].describe()
```

Out[92]:

	revenue_generated	revenue_realized	revenue_ratio
count	1.345900e+05	134590.000000	134590.000000
mean	1.537805e+04	12696.123256	1.403965
std	9.303604e+04	6928.108124	5.368782
min	6.500000e+03	2600.000000	1.000000
25%	9.900000e+03	7600.000000	1.000000
50%	1.350000e+04	11700.000000	1.000000
75%	1.800000e+04	15300.000000	1.000000
max	2.856000e+07	45220.000000	1000.000000

	revenue_generated	revenue_realized	revenue_ratio
count	1.345900e+05	134590.000000	134590.000000
mean	1.537805e+04	12696.123256	1.403965
std	9.303604e+04	6928.108124	5.368782
min	6.500000e+03	2600.000000	1.000000
25%	9.900000e+03	7600.000000	1.000000
50%	1.350000e+04	11700.000000	1.000000
75%	1.800000e+04	15300.000000	1.000000
max	2.856000e+07	45220.000000	1000.000000

```
In [93]: fact_booking[
    [
        "booking_id",
        "revenue_generated",
        "revenue_realized",
        "revenue_ratio"
    ]
].sort_values(
    "revenue_ratio",
    ascending=False
).head(20)
```

Out[93]:

		booking_id	revenue_generated	revenue_realized	revenue_ratio
	111	May012216559RT32	28560000	28560	1000.000000
	2	May012216558RT13	9100000	9100	1000.000000
	315	May012216562RT22	12600000	12600	1000.000000
	129176	Jul282216562RT26	10000000	12600	793.650794
	562	May012217559RT118	2000000	4420	452.488688
	134544	Jul312217564RT13	11050	4420	2.500000
	134547	Jul312217564RT16	11050	4420	2.500000
	22	May012216558RT25	12600	5040	2.500000
	24	May012216558RT27	13860	5544	2.500000
	28	May012216558RT211	12600	5040	2.500000
	34	May012216558RT217	15120	6048	2.500000
	134557	Jul312217564RT28	15300	6120	2.500000
	134561	Jul312217564RT212	15300	6120	2.500000
	134562	Jul312217564RT213	18360	7344	2.500000
	134563	Jul312217564RT214	15300	6120	2.500000
	134583	Jul312217564RT44	32300	12920	2.500000
	3	May012216558RT14	9100	3640	2.500000
	1	May012216558RT12	9100	3640	2.500000
	134587	Jul312217564RT48	32300	12920	2.500000
	134589	Jul312217564RT410	32300	12920	2.500000

```
In [94]: fact_booking["revenue_anomaly_flag"] = (  
    fact_booking["revenue_ratio"] > 10  
)
```

```
In [95]: fact_booking[  
    fact_booking["revenue_anomaly_flag"]  
][  
    [  
        "booking_id",  
        "revenue_generated",  
        "revenue_realized",  
        "revenue_ratio"  
    ]  
]
```

		booking_id	revenue_generated	revenue_realized	revenue_ratio
	2	May012216558RT13	9100000	9100	1000.000000
	111	May012216559RT32	28560000	28560	1000.000000
	315	May012216562RT22	12600000	12600	1000.000000
	562	May012217559RT118	2000000	4420	452.488688
	129176	Jul282216562RT26	10000000	12600	793.650794

```

In [96]: capacity_validation = {
    "missing_capacity":
        fact_aggregated_booking[
            "capacity"
        ].isna().sum(),

    "capacity_zero_or_negative":
        (
            fact_aggregated_booking["capacity"]
            <= 0
        ).sum(),

    "negative_successful_bookings":
        (
            fact_aggregated_booking[
                "successful_bookings"
            ] < 0
        ).sum(),

    "successful_bookings_over_capacity":
        (
            fact_aggregated_booking[
                "successful_bookings"
            ]
            >
            fact_aggregated_booking["capacity"]
        ).sum()
}

```

```

In [97]: capacity_validation

```

```

Out[97]: {'missing_capacity': np.int64(2),
          'capacity_zero_or_negative': np.int64(0),
          'negative_successful_bookings': np.int64(0),
          'successful_bookings_over_capacity': np.int64(6)}

```

```

In [98]: invalid_properties = fact_booking[
    ~fact_booking["property_id"].isin(
        dim_hotel["property_id"]
    )
]

print(

```

```
    "Invalid property IDs:",  
    len(invalid_properties)  
)
```

Invalid property IDs: 0

```
In [100... dim_room.columns
```

```
Out[100... Index(['room_id', 'room_class'], dtype='object')
```

```
In [101... invalid_rooms = fact_booking[  
    ~fact_booking["room_category"].isin(  
        dim_room["room_id"]  
    )  
]  
  
print(  
    "Invalid room IDs:",  
    len(invalid_rooms)  
)
```

Invalid room IDs: 0

```
In [102... validation_results = {  
  
    "duplicate_booking_ids":  
        fact_booking[  
            "booking_id"  
        ].duplicated().sum(),  
  
    "invalid_guest_counts":  
        (  
            fact_booking["no_guests"]  
            .fillna(1) <= 0  
        ).sum(),  
  
    "missing_guest_counts":  
        fact_booking[  
            "no_guests"  
        ].isna().sum(),  
  
    "negative_revenue":  
        (  
            fact_booking[  
                "revenue_realized"  
            ] < 0  
        ).sum(),  
  
    "booking_after_checkin":  
        (  
            fact_booking[  
                "booking_date"  
            ]  
            >  
            fact_booking[  
                "check_in_date"  
            ]  
        )
```

```

        ).sum(),

        "invalid_checkout":
        (
            fact_booking[
                "checkout_date"
            ]
            <=
            fact_booking[
                "check_in_date"
            ]
        ).sum(),

        "negative_lead_time":
        (
            fact_booking[
                "booking_lead_days"
            ] < 0
        ).sum(),

        "invalid_length_of_stay":
        (
            fact_booking[
                "length_of_stay"
            ] <= 0
        ).sum(),

        "invalid_property_ids":
            len(invalid_properties),

        "invalid_room_ids":
            len(invalid_rooms),

        "revenue_anomalies":
            fact_booking[
                "revenue_anomaly_flag"
            ].sum()
    }

```

In [103...

```

validation_df = pd.DataFrame(
    validation_results.items(),
    columns=[
        "validation_check",
        "failed_records"
    ]
)

validation_df

```

Out[103...

	validation_check	failed_records
0	duplicate_booking_ids	0
1	invalid_guest_counts	0
2	missing_guest_counts	12
3	negative_revenue	0
4	booking_after_checkin	0
5	invalid_checkout	0
6	negative_lead_time	0
7	invalid_length_of_stay	0
8	invalid_property_ids	0
9	invalid_room_ids	0
10	revenue_anomalies	5

In [104...

fact_booking.dtypes

Out[104...

```

booking_id                object
property_id               int64
booking_date              datetime64[ns]
check_in_date             datetime64[ns]
checkout_date            datetime64[ns]
no_guests                 Int64
room_category             object
booking_platform          object
ratings_given             float64
booking_status            object
revenue_generated         int64
revenue_realized          int64
booking_lead_days         int64
lead_time_group           string[python]
length_of_stay            int64
guest_count_available     bool
revenue_ratio             float64
revenue_anomaly_flag      bool
dtype: object

```

In [105...

validation_df

Out[105...

	validation_check	failed_records
0	duplicate_booking_ids	0
1	invalid_guest_counts	0
2	missing_guest_counts	12
3	negative_revenue	0
4	booking_after_checkin	0
5	invalid_checkout	0
6	negative_lead_time	0
7	invalid_length_of_stay	0
8	invalid_property_ids	0
9	invalid_room_ids	0
10	revenue_anomalies	5

In [107...

fact_booking.dtypes

Out[107...

```

booking_id           object
property_id          int64
booking_date         datetime64[ns]
check_in_date        datetime64[ns]
checkout_date        datetime64[ns]
no_guests            Int64
room_category        object
booking_platform      object
ratings_given        float64
booking_status       object
revenue_generated    int64
revenue_realized     int64
booking_lead_days    int64
lead_time_group      string[python]
length_of_stay       int64
guest_count_available bool
revenue_ratio        float64
revenue_anomaly_flag bool
dtype: object

```

In [108...

capacity_validation

Out[108...

```

{'missing_capacity': np.int64(2),
 'capacity_zero_or_negative': np.int64(0),
 'negative_successful_bookings': np.int64(0),
 'successful_bookings_over_capacity': np.int64(6)}

```

In [110...

```

fact_booking[
    fact_booking["revenue_anomaly_flag"]
][
    [

```

```

        "booking_id",
        "revenue_generated",
        "revenue_realized",
        "revenue_ratio"
    ]
].head(20)

```

Out[110...

		booking_id	revenue_generated	revenue_realized	revenue_ratio
	2	May012216558RT13	9100000	9100	1000.000000
	111	May012216559RT32	28560000	28560	1000.000000
	315	May012216562RT22	12600000	12600	1000.000000
	562	May012217559RT118	2000000	4420	452.488688
	129176	Jul282216562RT26	10000000	12600	793.650794

In [111...

```

revenue_anomalies = fact_booking[
    fact_booking["revenue_anomaly_flag"]
]

revenue_anomalies[
    [
        "booking_id",
        "property_id",
        "room_category",
        "booking_platform",
        "booking_status",
        "revenue_generated",
        "revenue_realized",
        "no_guests",
        "length_of_stay"
    ]
]

```

Out[111...

		booking_id	property_id	room_category	booking_platform	booking_status
	2	May012216558RT13	16558	RT1	logtrip	Checked Out
	111	May012216559RT32	16559	RT3	direct online	Checked Out
	315	May012216562RT22	16562	RT2	direct offline	Checked Out
	562	May012217559RT118	17559	RT1	others	Canceled
	129176	Jul282216562RT26	16562	RT2	direct online	Checked Out

In [112...

```

fact_booking.groupby(
    "room_category"
)[
    [
        "revenue_generated",
        "revenue_realized"
    ]
]

```

```
]
].median()
```

Out[112... revenue_generated revenue_realized

room_category		
RT1	9750.0	9100.0
RT2	13500.0	12600.0
RT3	18000.0	16800.0
RT4	28500.0	26600.0

```
In [113... fact_booking.groupby(
    "room_category"
)["revenue_generated"].describe()
```

	count	mean	std	min	25%	50%	75%
room_category							
RT1	38446.0	9744.187822	47497.368234	6500.0	7800.0	9750.0	11050.0
RT2	49505.0	13766.709625	72259.700138	9000.0	12600.0	13500.0	15300.0
RT3	30566.0	18717.832886	163299.549586	12000.0	15600.0	18000.0	20400.0
RT4	16073.0	27465.751260	6310.852167	19000.0	19000.0	28500.0	32300.0

```
In [115... fact_booking[
    "revenue_generated_clean"
] = fact_booking[
    "revenue_generated"
].copy()
```

```
In [116... factor_1000_mask = (
    fact_booking["revenue_ratio"]
    .between(999, 1001)
)
```

```
In [117... fact_booking.loc[
    factor_1000_mask,
    "revenue_generated_clean"
] = (
    fact_booking.loc[
        factor_1000_mask,
        "revenue_generated"
    ] / 1000
)
```

```
In [118... # capacity
capacity_lookup = (
    fact_aggregated_booking
```

```
.groupby([
    "property_id",
    "room_category"
])["capacity"]
.transform("median")
)
```

```
In [119... fact_aggregated_booking[
    "capacity_clean"
] = fact_aggregated_booking[
    "capacity"
]
```

```
In [120... fact_aggregated_booking[
    "capacity_clean"
] = (
    fact_aggregated_booking[
        "capacity_clean"
    ]
    .fillna(capacity_lookup)
)
```

```
In [121... fact_aggregated_booking[
    "capacity_imputed"
] = (
    fact_aggregated_booking[
        "capacity"
    ].isna()
)
```

```
In [122... fact_aggregated_booking[
    "capacity_clean"
].isna().sum()
```

```
Out[122... np.int64(0)
```

```
In [ ]:
```